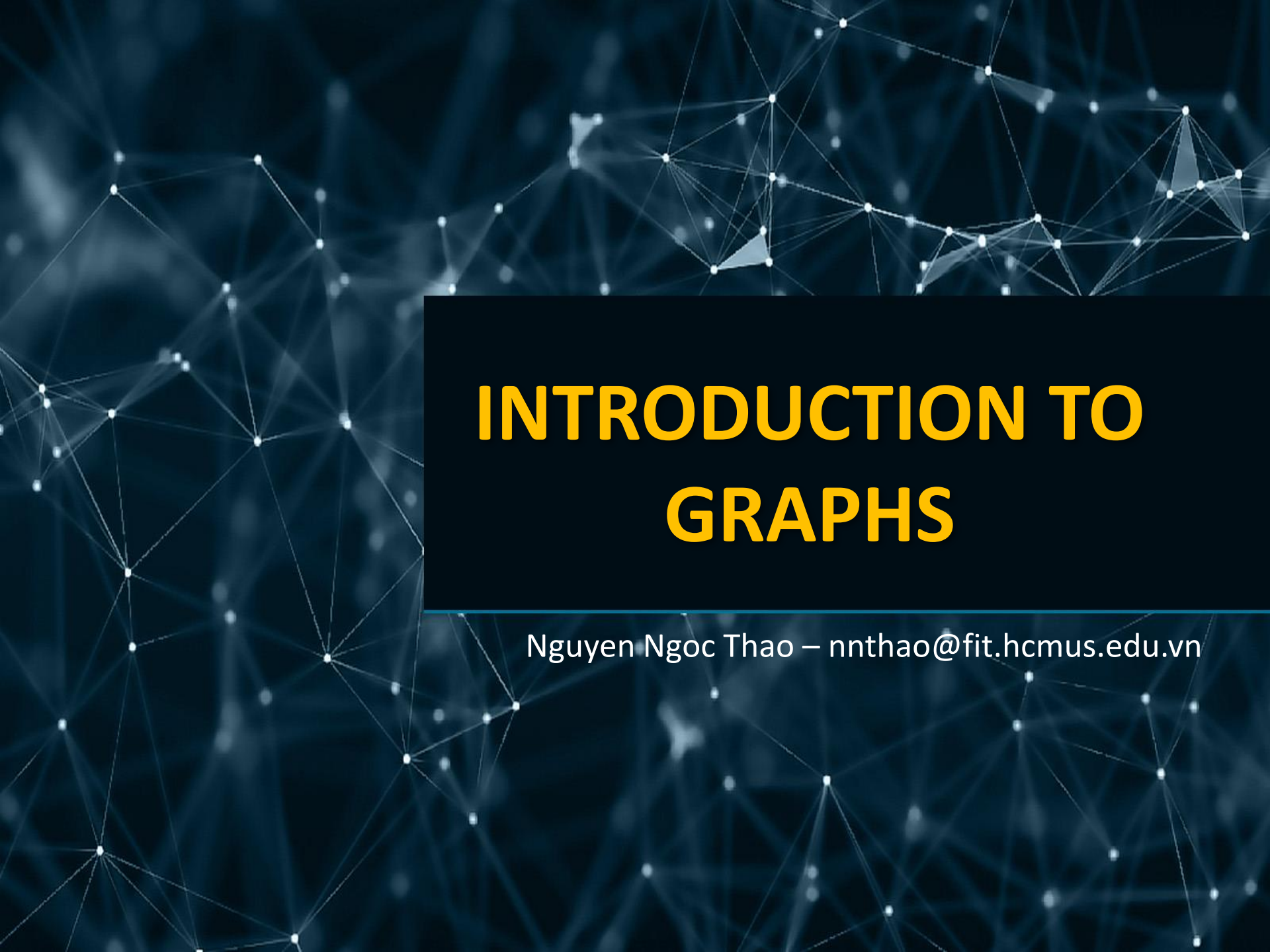


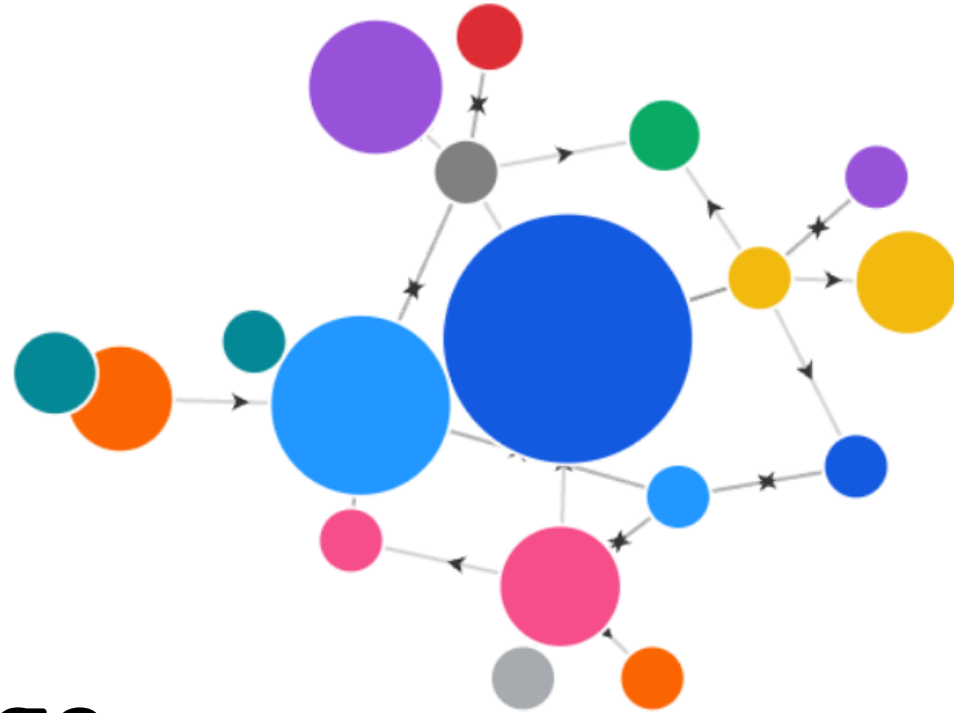


INTRODUCTION TO GRAPHS

Nguyen Ngoc Thao – nnthao@fit.hcmus.edu.vn

Outline

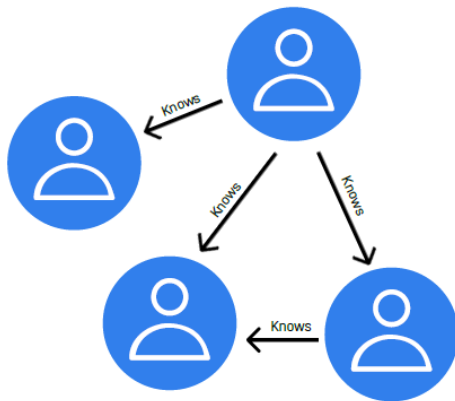
- Knowledge representation with graphs
- Graph mining tasks
- Graph data: Sources and tools



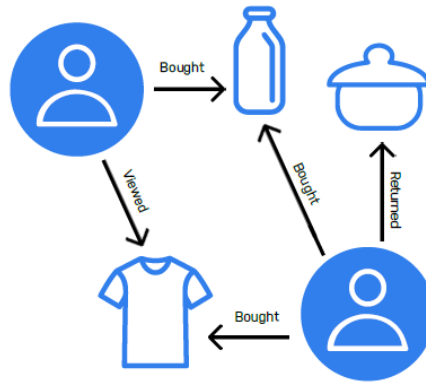
Knowledge
representation
with graphs

Graphs for data representation

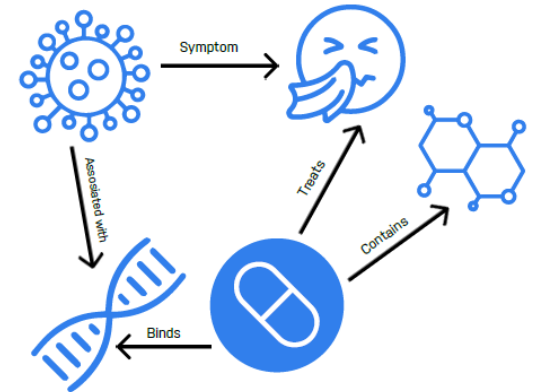
- **Graphs** enable the **effective representation and analysis** of **complex relationships** among objects.
- Therefore, it has gained **increasing importance** in **machine learning and data mining**.



Networks of People



Transaction Networks



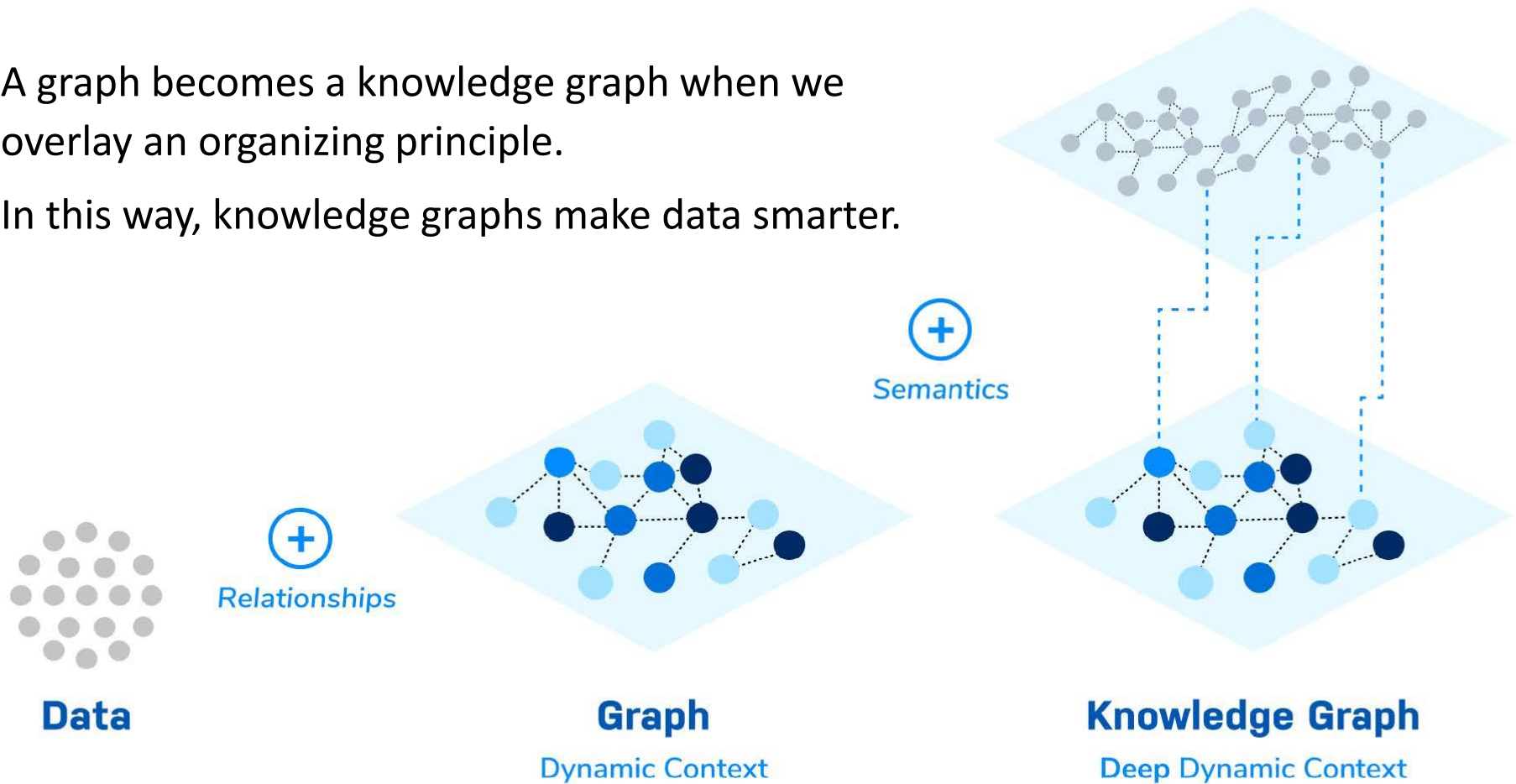
Knowledge Networks

Examples of graph representations. Everything is naturally connected.

From data to knowledge graphs

A graph becomes a knowledge graph when we overlay an organizing principle.

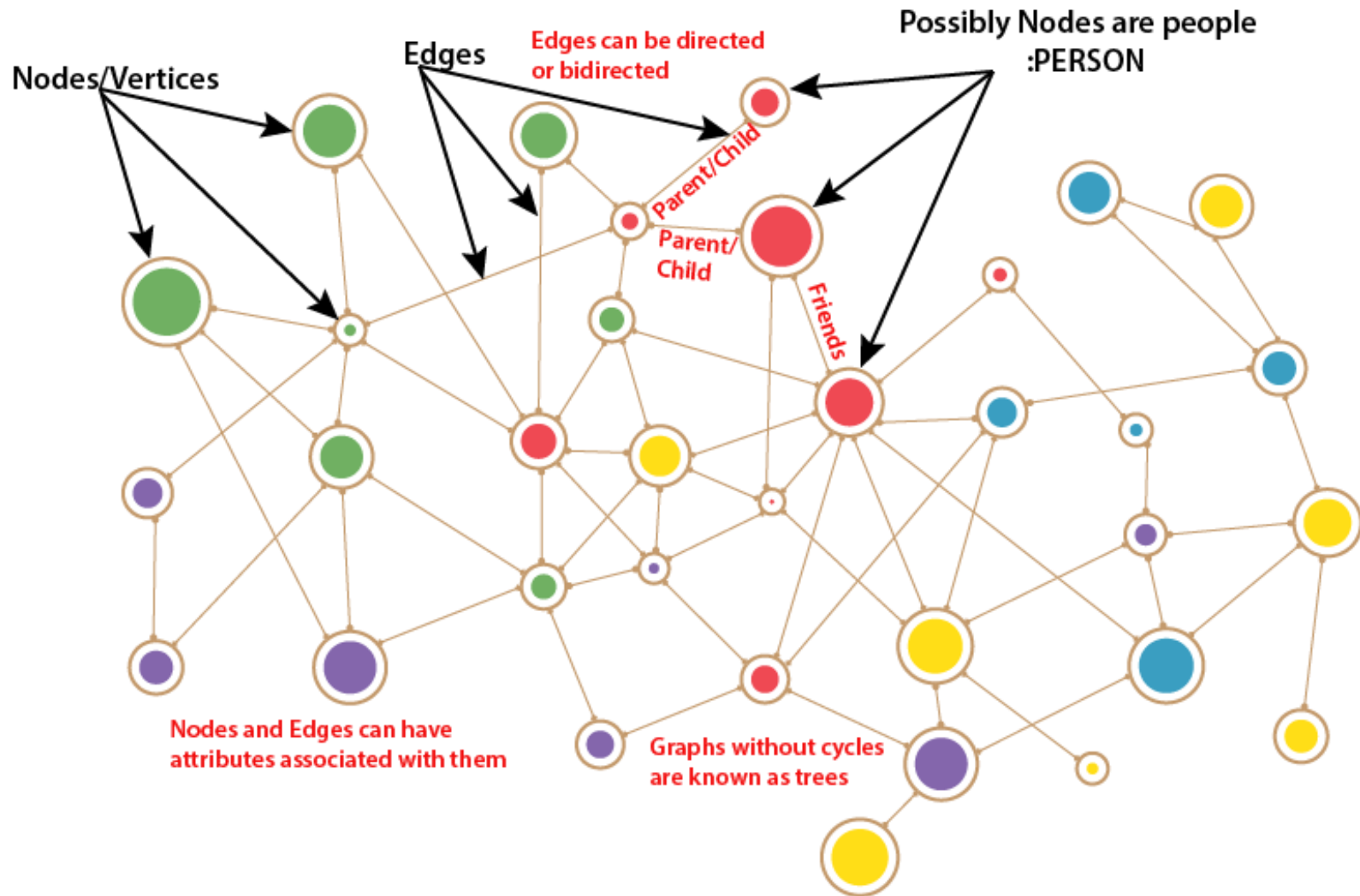
In this way, knowledge graphs make data smarter.



What is a knowledge graph?

- A **knowledge graph** is a **structured representation of facts**, including **entities, relationships, and semantic descriptions**.
 - Entities can be real-world objects and abstract concepts.
 - Relationships represent the relation between entities.
- Property (or attributed) graphs: entities and relations may have several properties.

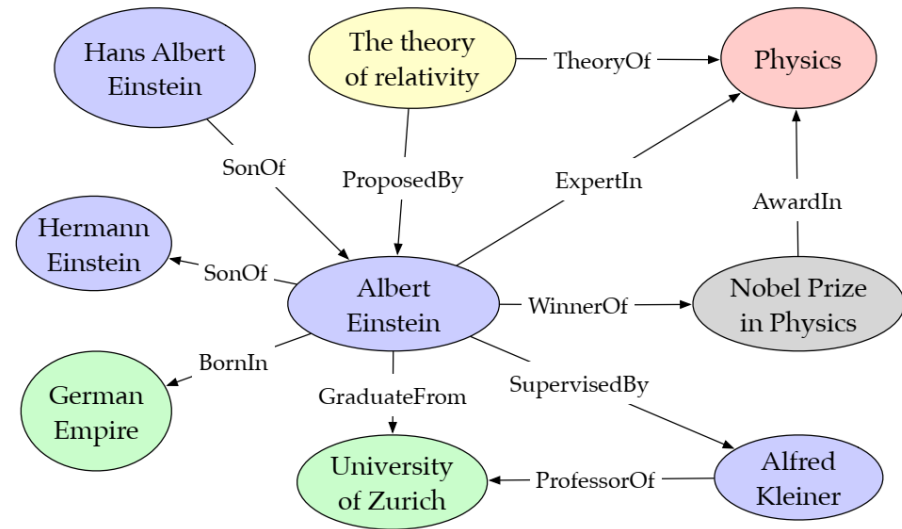
Knowledge graphs: Terminologies



What is a knowledge graph?

(Albert Einstein, **BornIn**, German Empire)
(Albert Einstein, **SonOf**, Hermann Einstein)
(Albert Einstein, **GraduateFrom**, University of Zurich)
(Albert Einstein, **WinnerOf**, Nobel Prize in Physics)
(Albert Einstein, **ExpertIn**, Physics)
(Nobel Prize in Physics, **AwardIn**, Physics)
(The theory of relativity, **TheoryOf**, Physics)
(Albert Einstein, **SupervisedBy**, Alfred Kleiner)
(Alfred Kleiner, **ProfessorOf**, University of Zurich)
(The theory of relativity, **ProposedBy**, Albert Einstein)
(Hans Albert Einstein, **SonOf**, Albert Einstein)

(a) Factual triples in knowledge base.

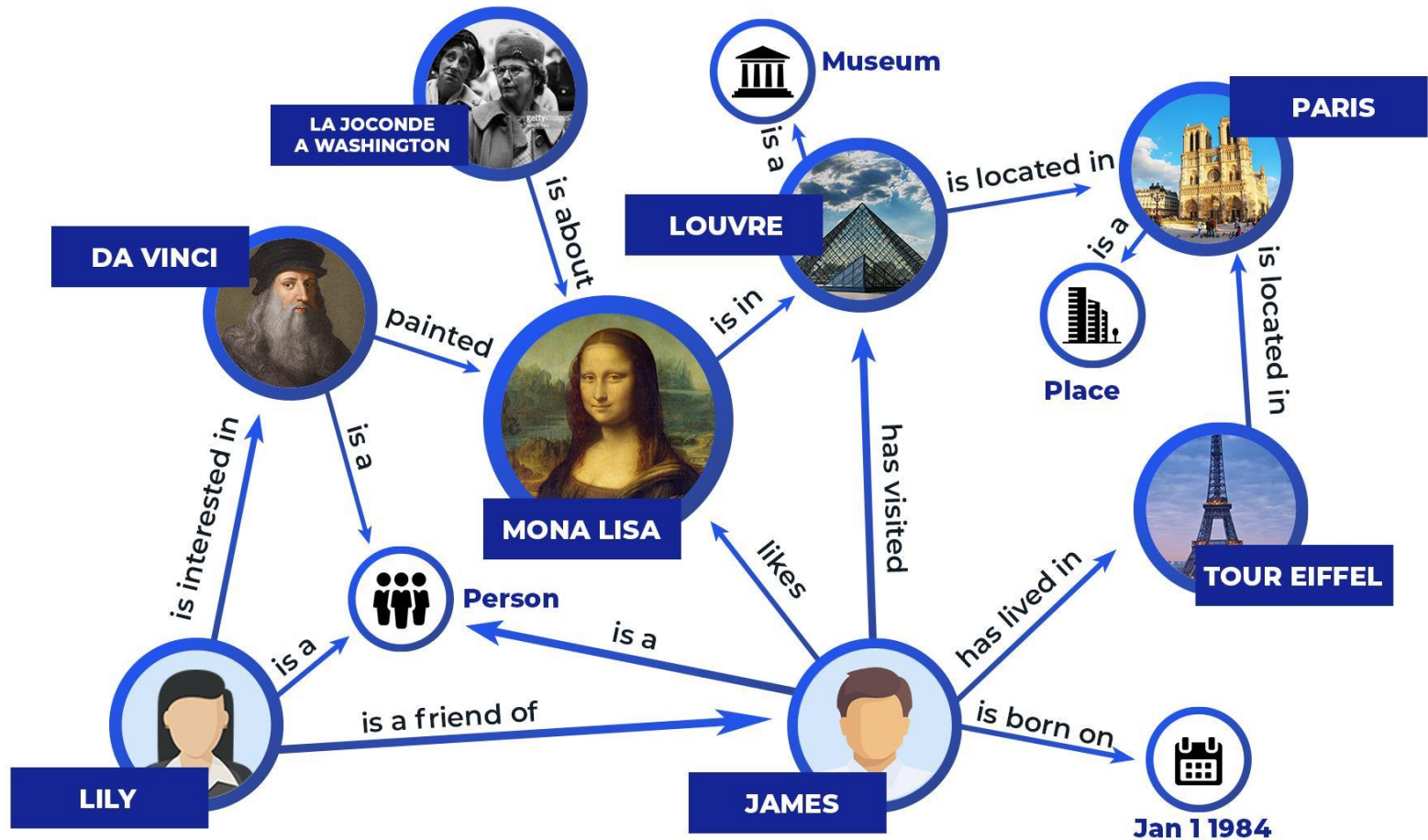


(b) Entities and relations in knowledge graph.

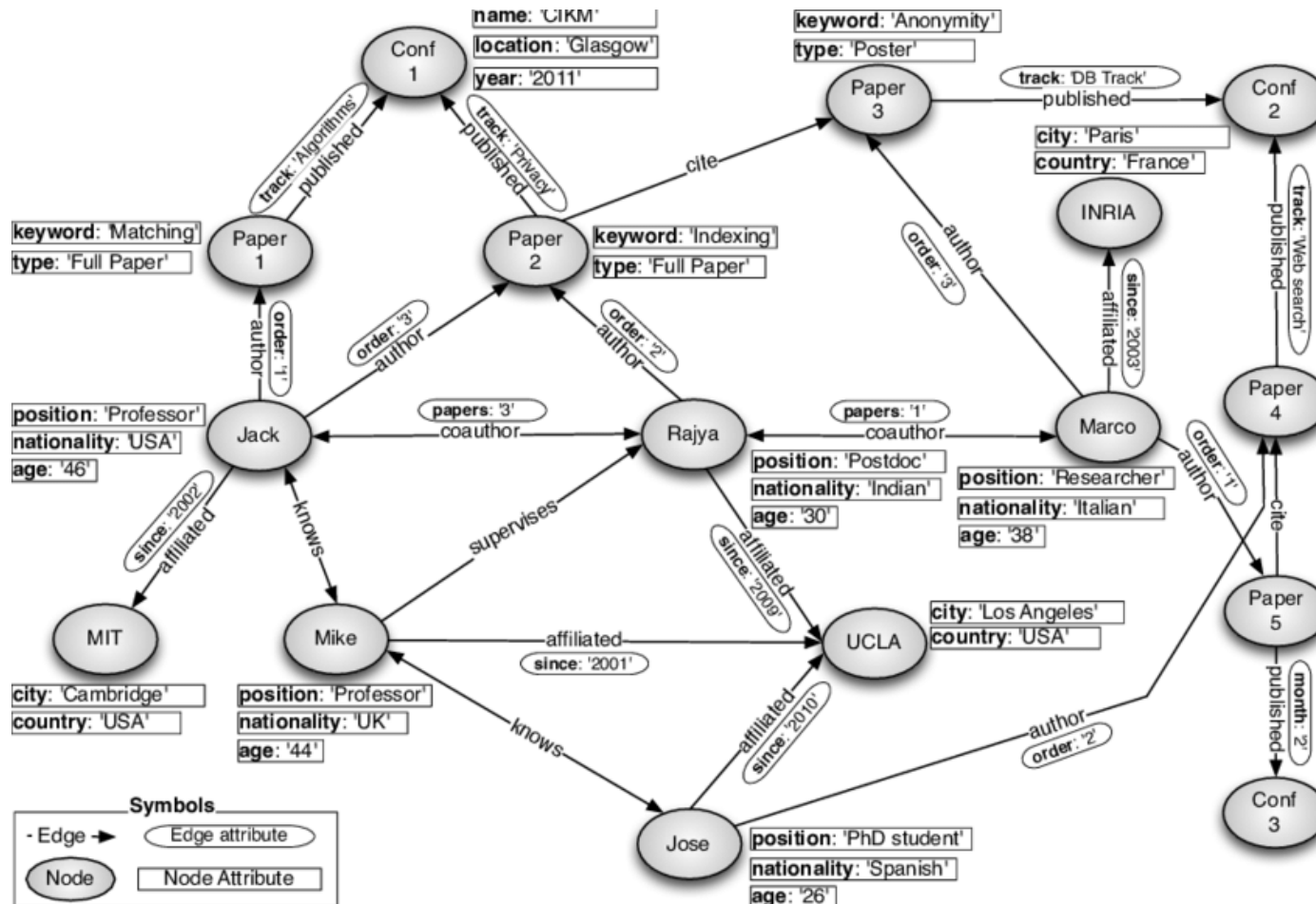
An example of knowledge base and knowledge graph.

(a) Knowledge can be expressed in a factual triple in the form of (head; relation; tail) or (subject; predicate; object). (b) It can also be represented as a directed graph with nodes as entities and edges as relations.

Knowledge graphs: Examples

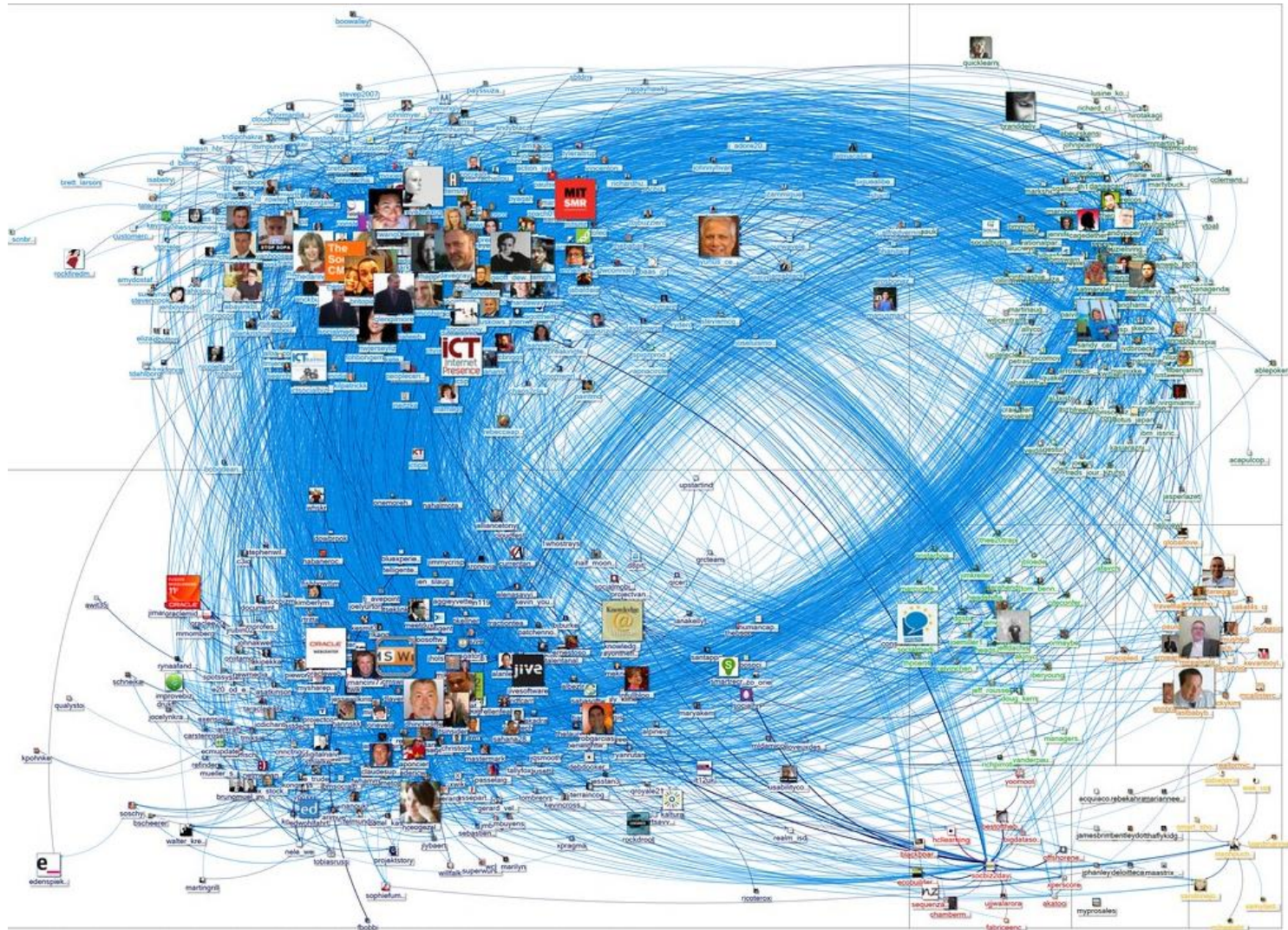


Knowledge graphs: Examples



An example of attributed graph. Both nodes and edges have attributes.

Case studies: Social network

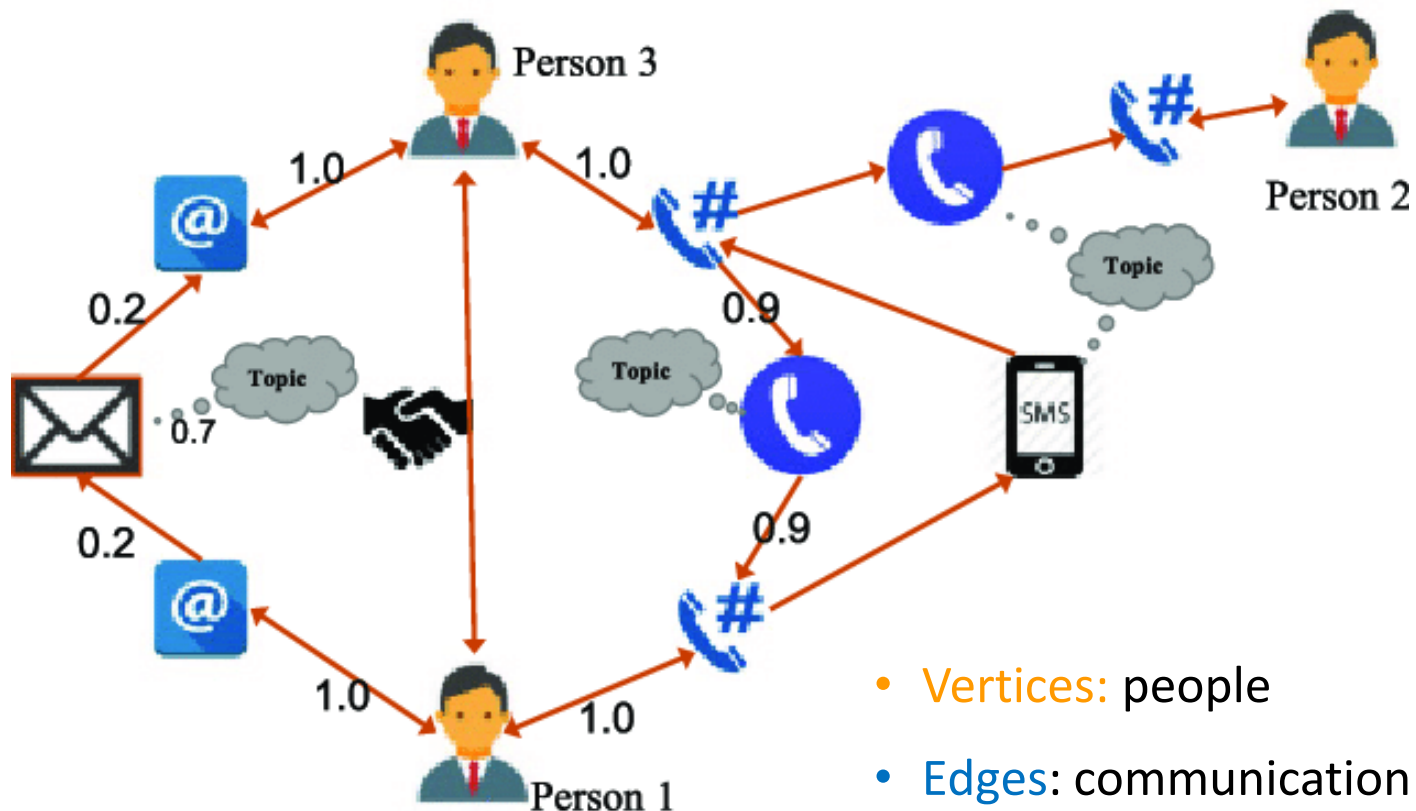


- Vertices: users

- Edges: connections

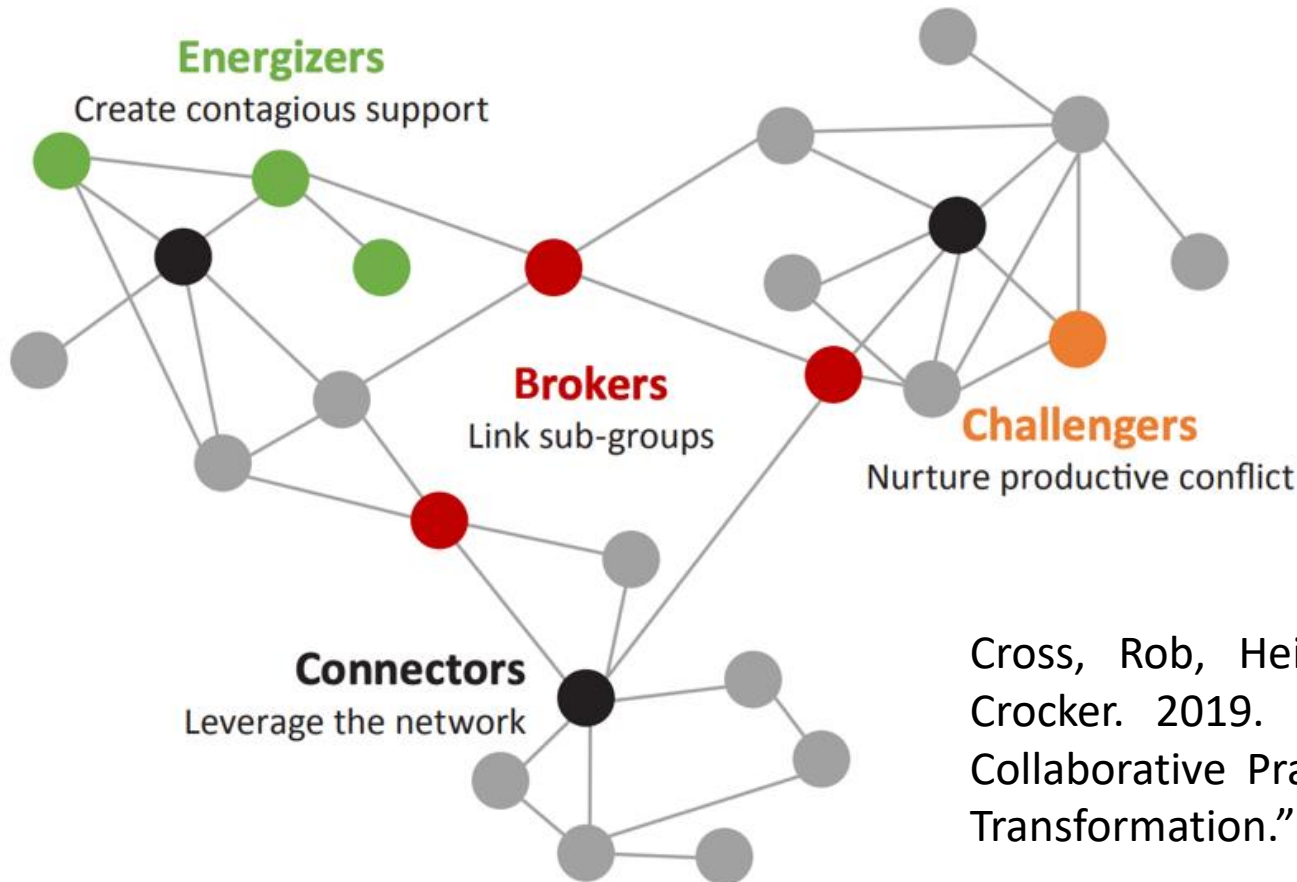
Case studies: Communication network

- Identify collaborative networks over email, phone, calendar, and many more digital communication means



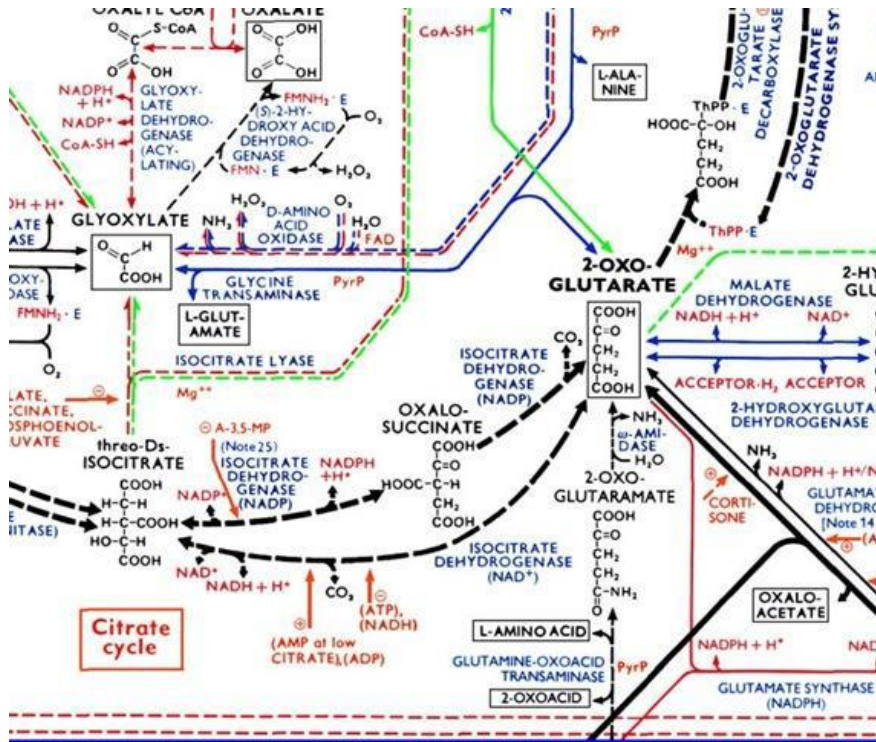
Case studies: Organization graph

- Explain the organizational or inter-organizational dynamics

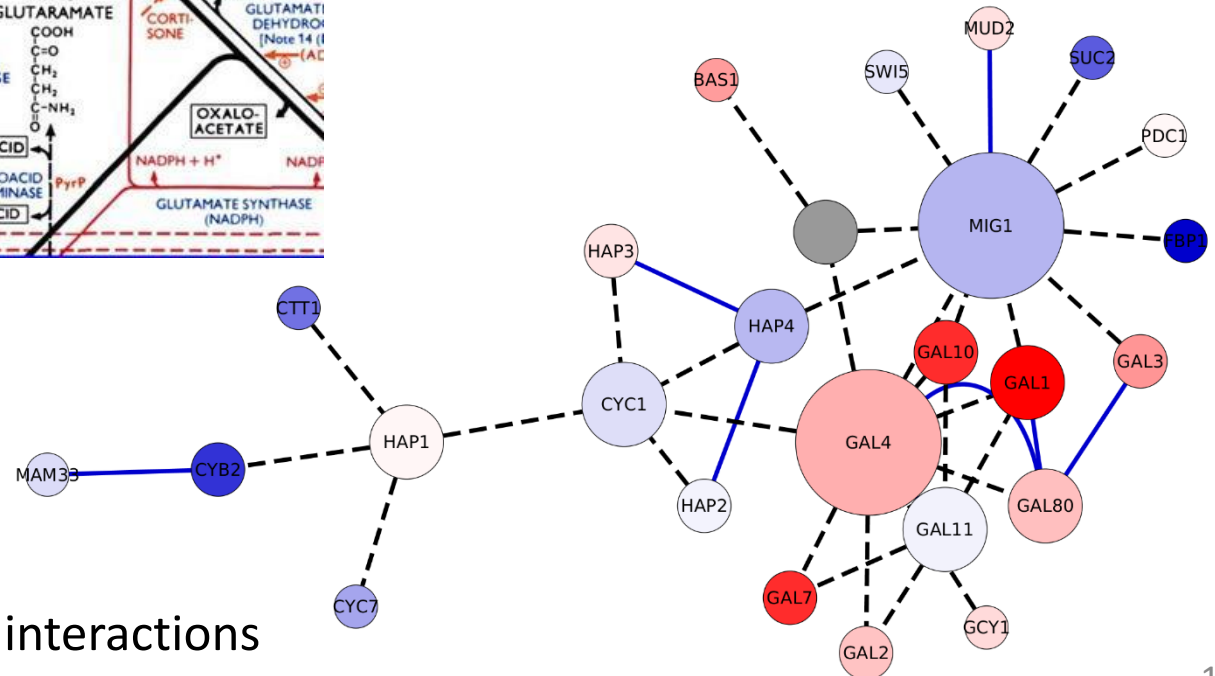


Cross, Rob, Heidi Gardner, and Alia Crocker. 2019. "Networks for Agility: Collaborative Practices Critical to Agile Transformation." Connected Commons.

Case studies: Biology network



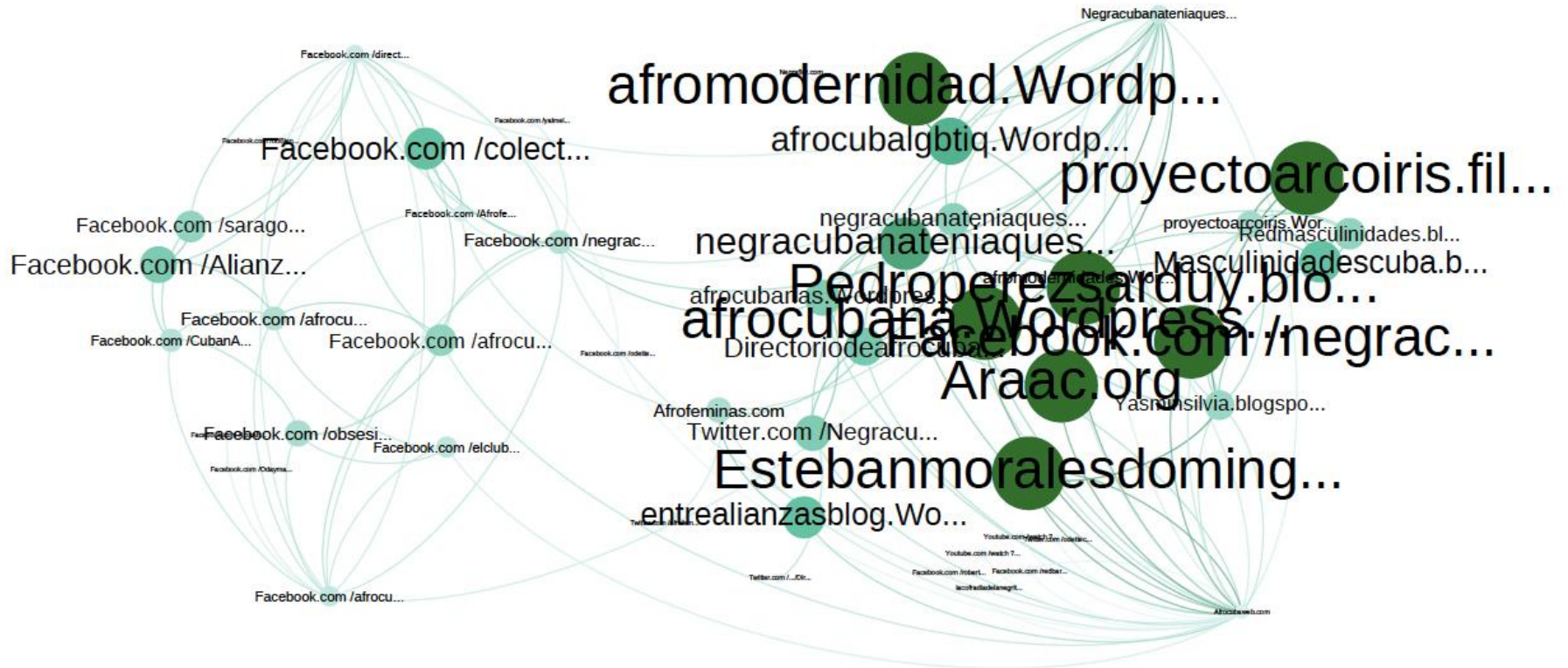
- **Vertices:** metabolites, enzymes
- **Edges:** chemical reactions



- **Vertices:** proteins
- **Edges:** interactions

Case studies: WWW

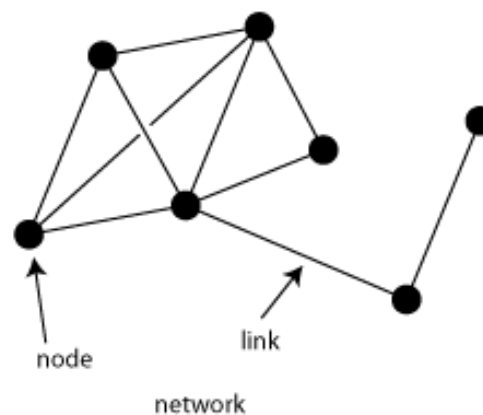
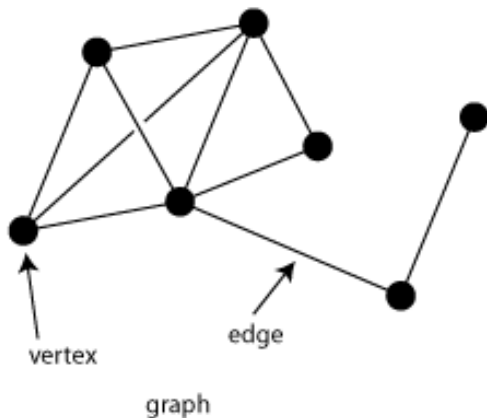
- WWW is considered the biggest network in the virtual world.



- **Vertices:** web pages
- **Edges:** hyperlinks

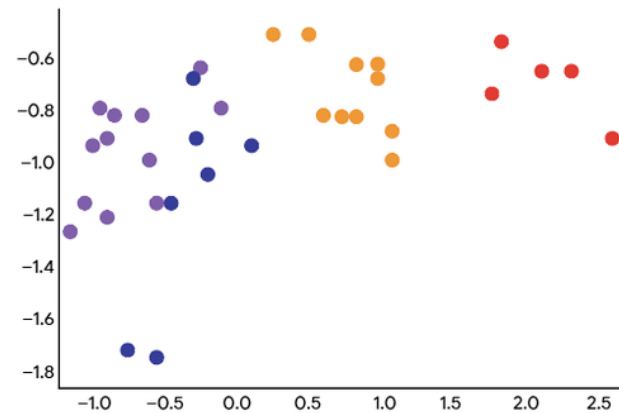
Graphs and Networks

- **Networks** are used to represent the **natural relationships among objects** in the real world.
 - E.g., WWW is a network of web pages linked by URLs, society is a network of individuals linked by family, friendship or professional ties.
- **Graphs** are mathematical representations of these networks.



NETWORK SCIENCE	GRAPH THEORY
Network	Graph
Node	Vertex
Link	Edge

Graph Mining Tasks

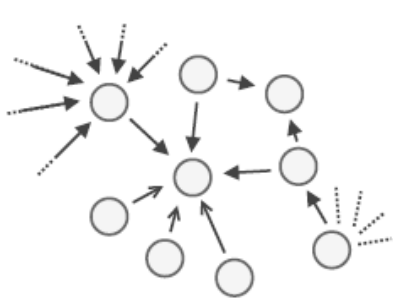


Graph mining: An introduction

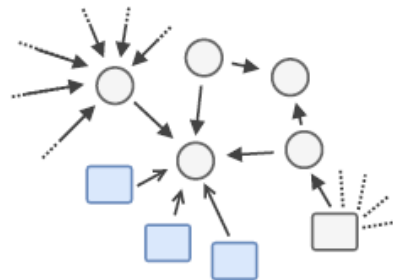
- **Graph mining** uses extensive mining techniques to **find a pattern or relationship** in the real-world **collection of graphs**.
- A graph mining task should address the following points
 - **What are the structural properties of the graph?**
 - **Which kinds of interactions take place in the graph?**
- **More challenging** than mining itemsets in transactional data
 - A graph is composed of multiple connected components.
 - Thus, simply relying the co-occurrence of items is insufficient.
- This requires **novel techniques for storage and computation**.
 - Graphs varies in terms of size, density, compactness and dynamics.

Mining tasks: Network analysis

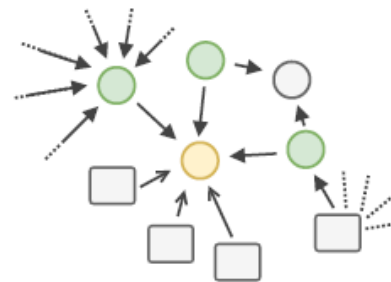
- Use both **graphical and statistical procedures** to **analyze the structure of a network**.



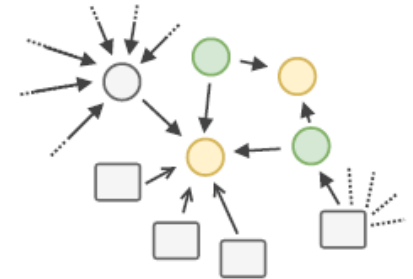
RAW Graph



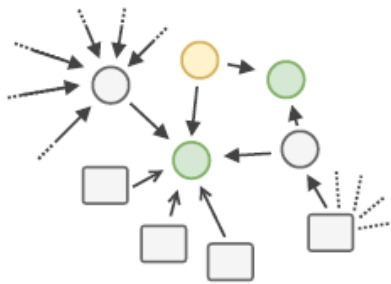
Co-Authorship
(Collaboration)



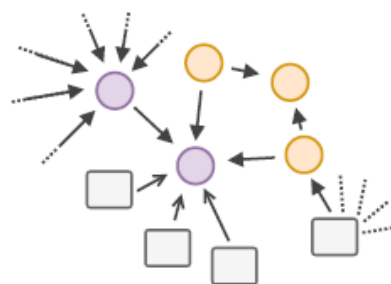
Citations



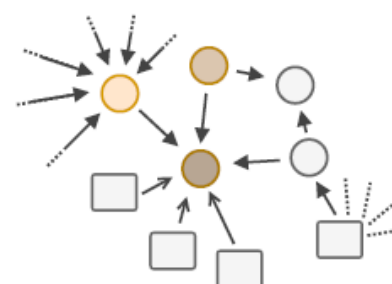
Co-Citations



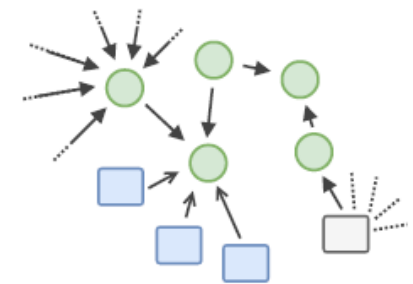
Bibliographical
Coupling



Topics



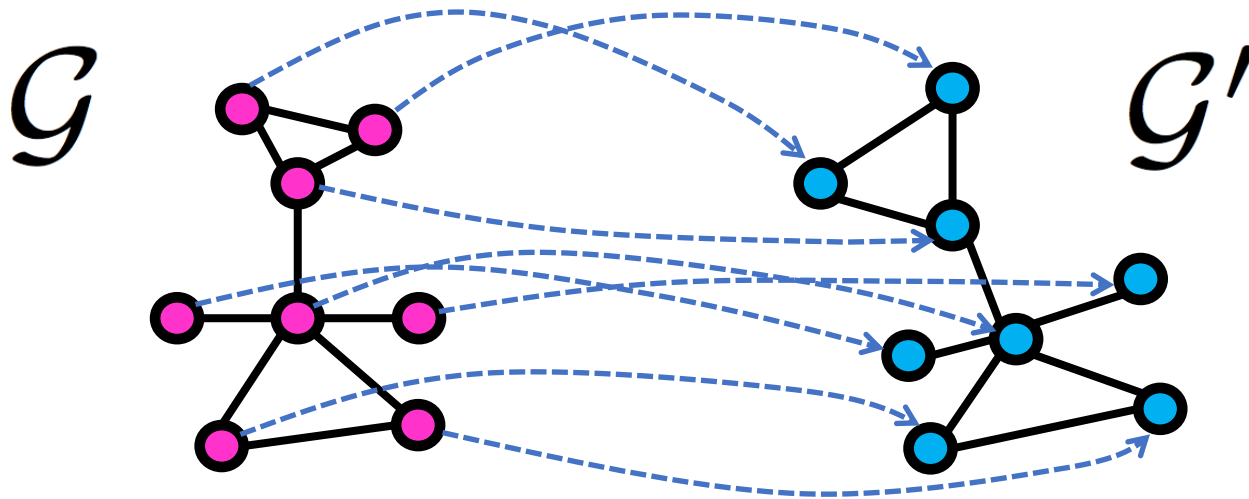
Co-Words



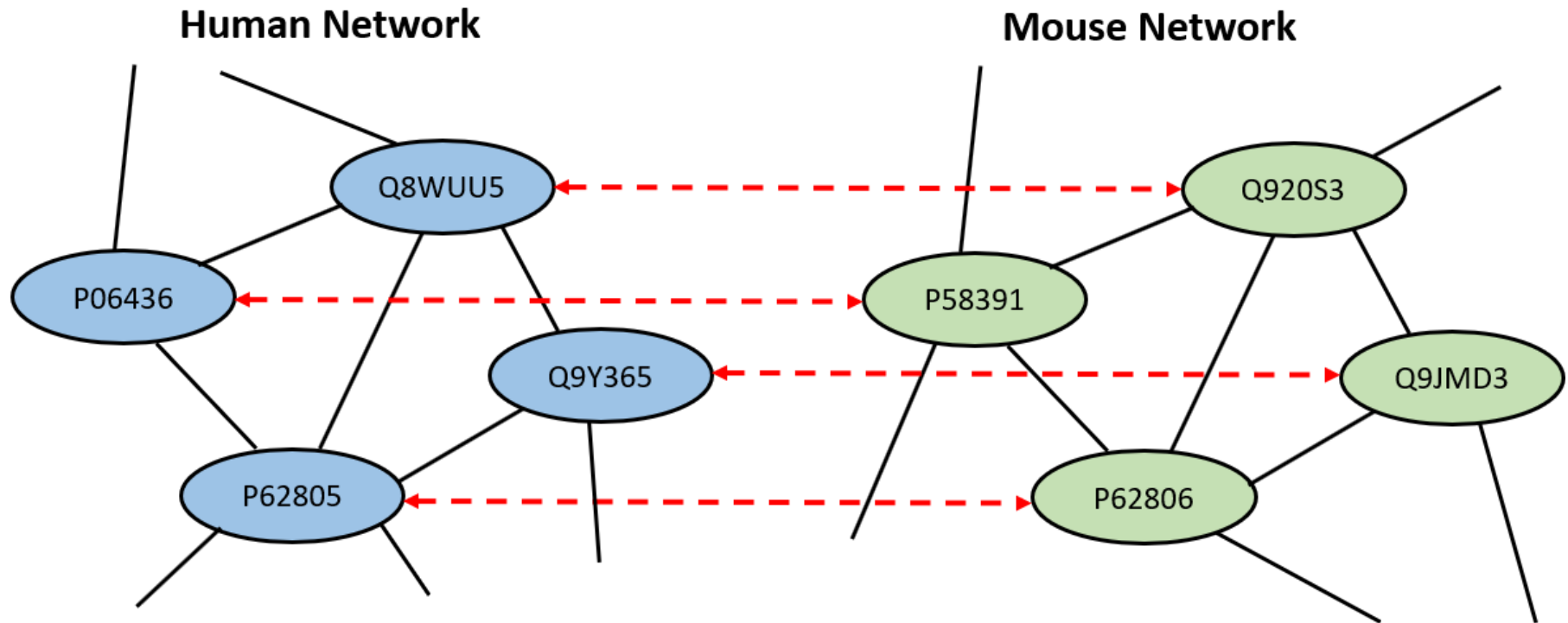
Heterogeneous
Networks

Mining tasks: Graph matching

- Find the structural similarities between two graphs.
 - Exact graph matching is known as the graph isomorphism problem.



Mining tasks: Graph matching

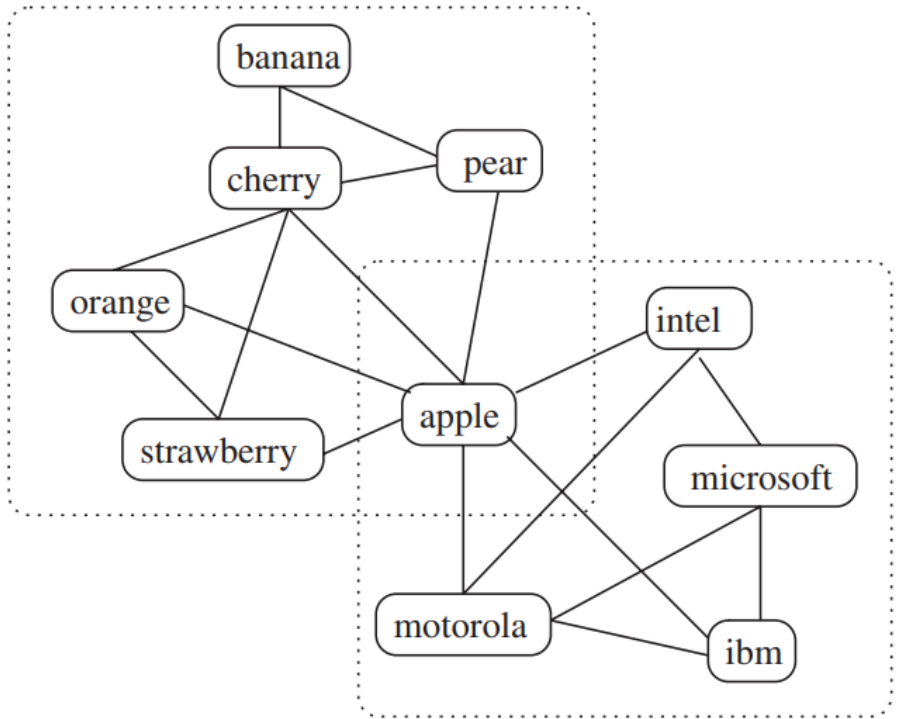
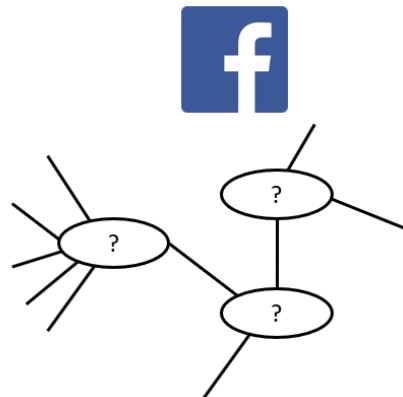
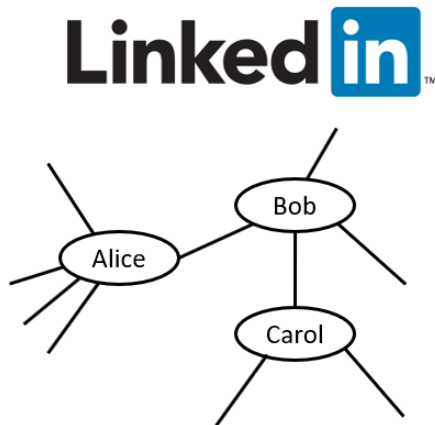


Find proteins of similar functionalities on different species
based on their interaction network

Kazemi, Ehsan, et al. "PROPER: global protein interaction network alignment through percolation matching." BMC bioinformatics 17.1 (2016): 1-16.

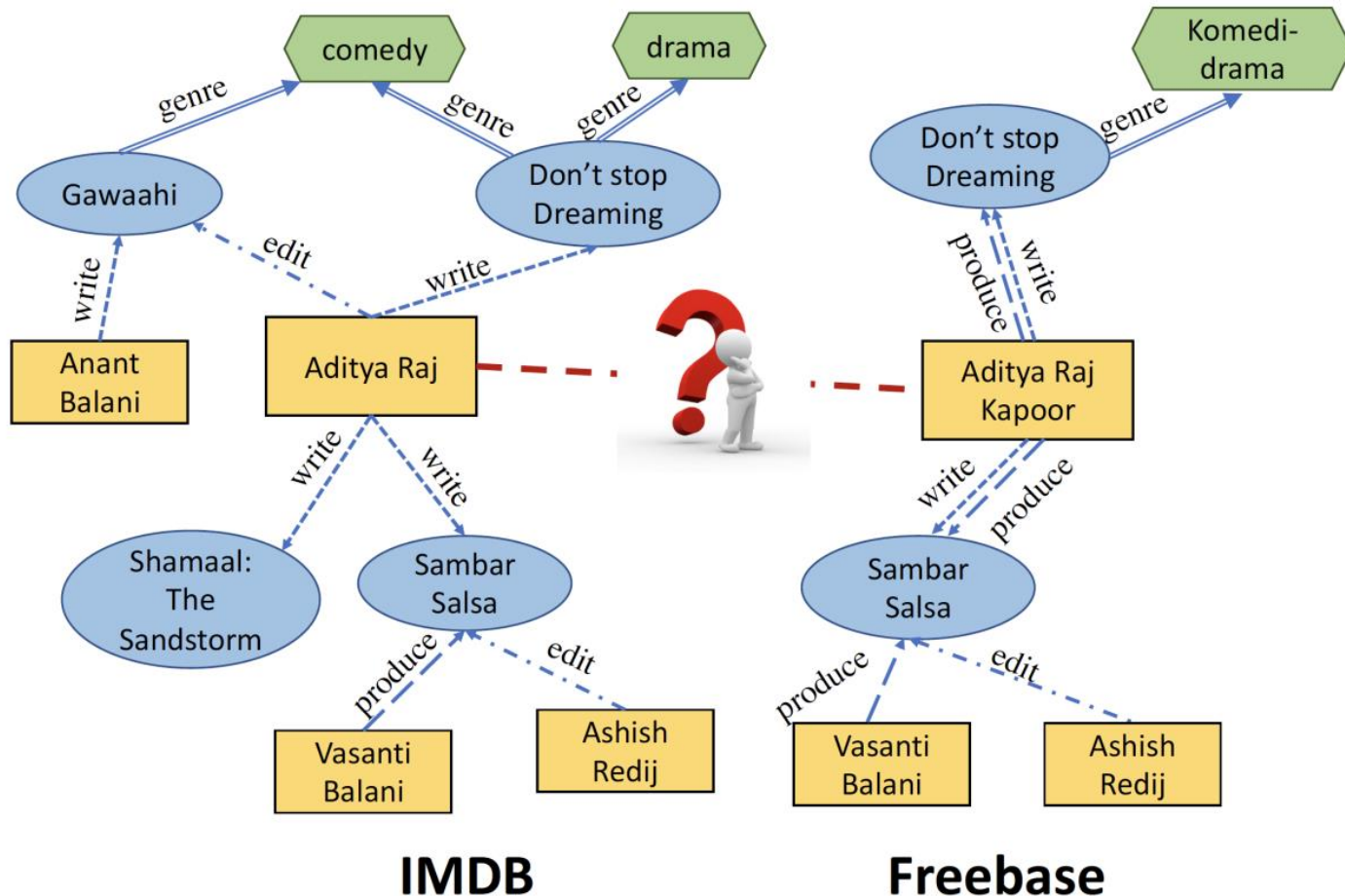
Mining tasks: Graph matching

Determine semantic classes



Use network topology to reveal hidden identities across social networks

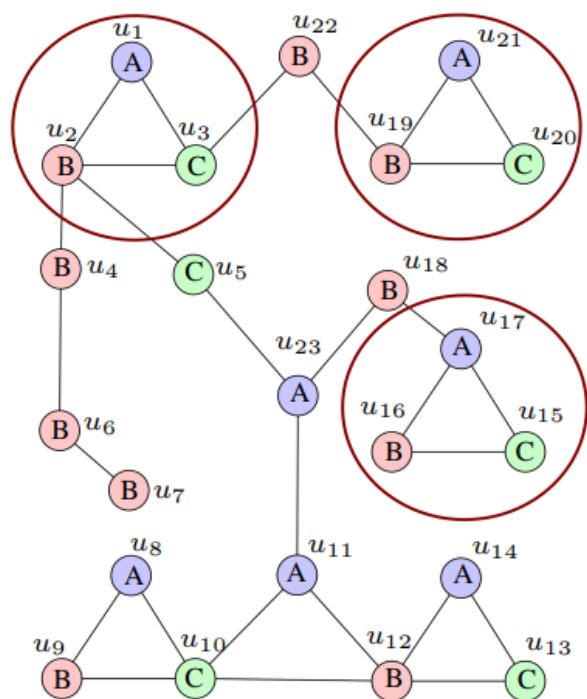
Mining tasks: Graph matching



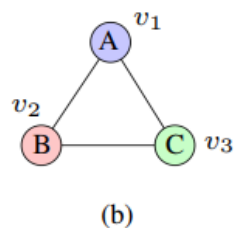
Automatically perform entity alignment, or determining which elements of different graphs refer to the same entities.

Mining tasks: Frequent patterns

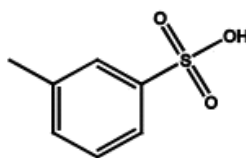
Input Graph G



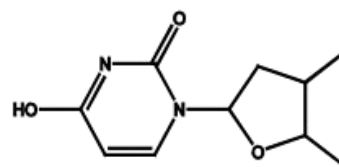
Subgraph S



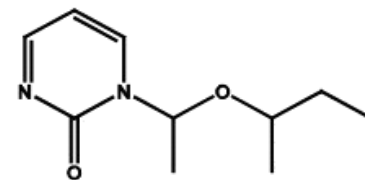
- A (sub)graph is **frequent** in a given dataset if its **support** is **no less than** a minimum support threshold.



(A)



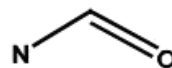
(B)



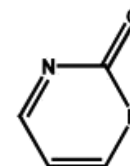
(C)

Frequent chemical bonds
(minsup is 2)

(1)



(2)

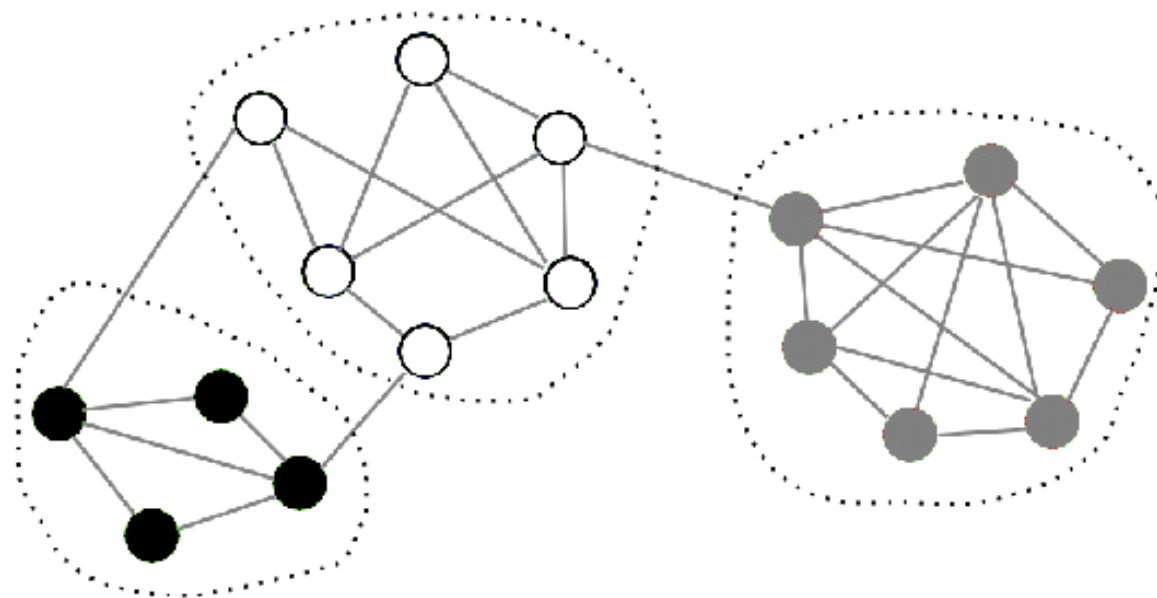


Mining tasks: Frequent patterns

<i>Algorithms</i>	<i>Input type</i>	<i>Dynamicity of Graphs</i>	<i>Result type</i>	<i>Algorithmic approach</i>	<i>Algorithmic design</i>	<i>Graph representation</i>
AGM [15]	undirected /directed graph set	static	all induced frequent subgraphs	apriori	serial	adjacency matrix
FSG [16]	undirected graph set	static	all frequent subgraphs	apriori	serial	adjacency list
FFSM [14]	undirected graph set	static	all frequent subgraphs	pattern growth	serial	CAM
MOFA [18]	undirected /directed graph set	static	all frequent subgraphs	pattern growth	serial	adjacency list
gSpan [17]	undirected graph set	static	all frequent subgraphs	pattern growth	serial	min DFS code
CloseGraph [13]	undirected graph set	static	all closed frequent subgraphs	pattern growth	serial	min DFS code
GERM [29]	undirected single graph	dynamic	all frequent subgraphs	pattern growth	serial	min DFS code

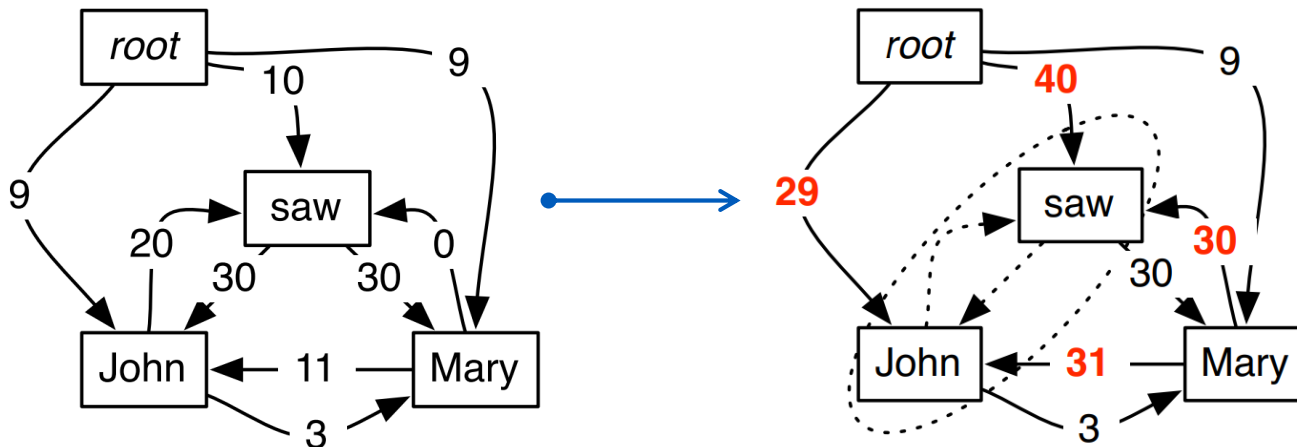
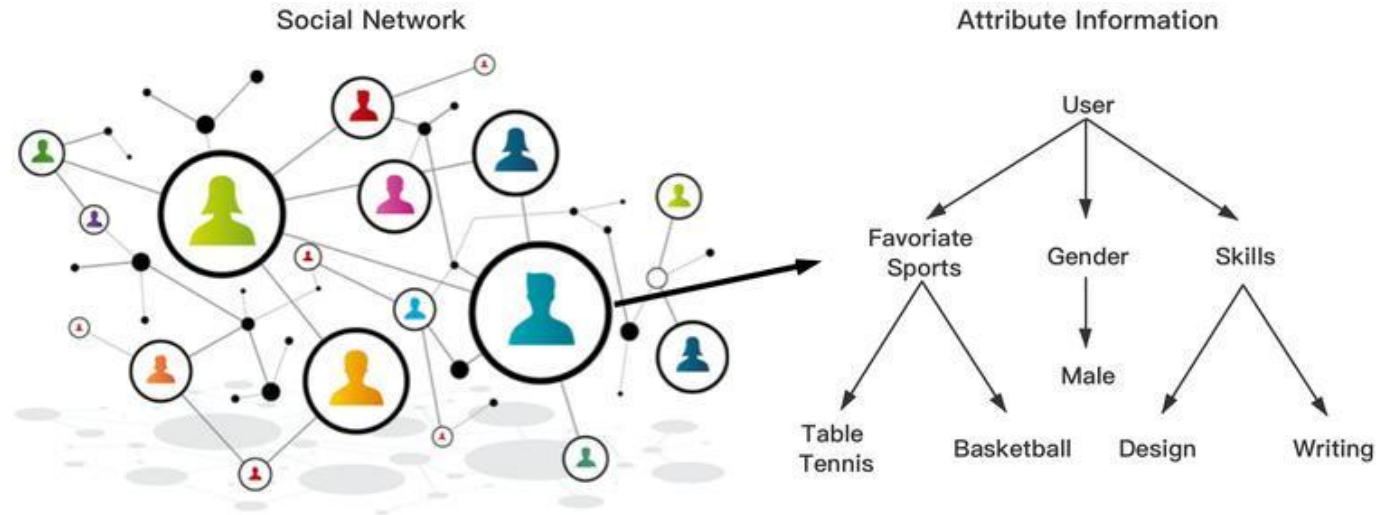
Mining tasks: Graph clustering

- Group the nodes of the graph into clusters, while taking into account the edge structure of the graph.
 - There are many edges within each cluster and few between clusters.



Mining tasks: Graph clustering

Community detection in attributed graphs



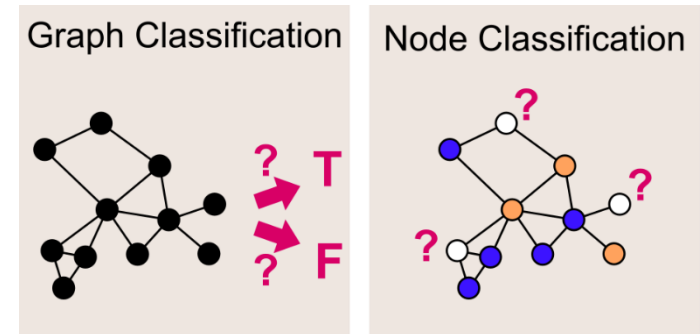
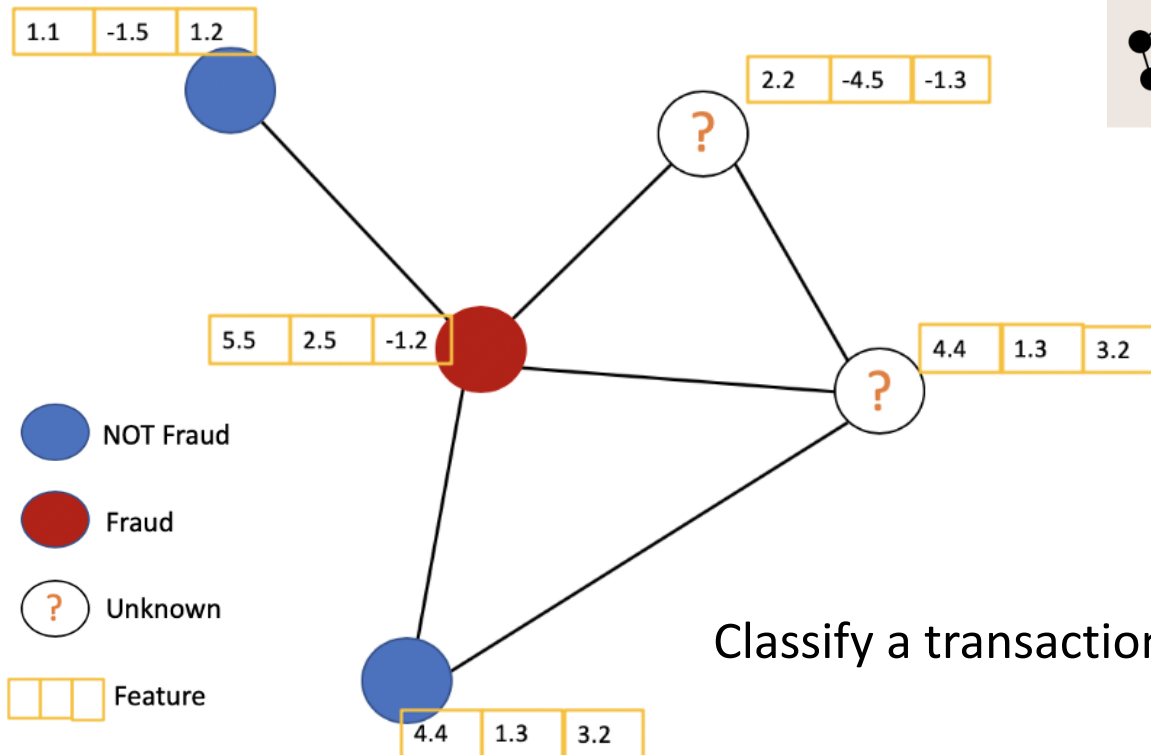
Graph summarization:
reduce the
graph's complexity

Mining tasks: Graph clustering

Method	Category	Clustering Type	Objective Function	Additional Features	Complexity
Symmetrization [54]	1 st	All			$\mathcal{O}(\sum_i D(i))$
Network Embedding [56]	1 st , 2 nd	All	Directed modularity	Laplacian matrix	Selected algorithm
Random Walk Sim [64]	1 st	All			Selected algorithm
Bipartite Modularity [66]	1 st , 2 nd	All	Bipartite modularity	Directed to Bipartite	Modularity Opt.
MAGA method [67]	1 st , 2 nd	All	Bipartite modularity	Genetic algorithm (GA)	Complexity of GA
Semi-supervised Learning [57]	1 st , 2 nd	All	Modified modularity		Spectral Opt.
Directed Modularity [50, 58]	1 st , 2 nd	All	Directed modularity		Spectral Opt.
LinkRank [55]	2 nd	All	Directed modularity		Modularity Opt.
DGRN [70, 71]	2 nd	All	Directed modularity		Spectral Opt.
Overlapping Modularity [73]	2 nd	Density-based	Dir.-Overl. modularity	Genetic Algorithm	$\mathcal{O}(C n^2)$
Local Modularity [74]	2 nd	Local cohesive groups	Local modularity		Modularity Opt.
Directed Laplacian [75]	2 nd	Density-based	Normalized Cuts		Spectral Opt.
Directed Normalized Laplacian [61]	2 nd	Density-based	Normalized Cuts		Spectral Opt.
Multiple Views Spectral Clus. [76]	2 nd	Density-based	Multiple Cuts	Multiple Views	Spectral Opt.
Weighted Cuts [59]	2 nd	Density-based	Weighted cuts		Spectral Opt.
Attraction and Repulsion [77]	1 st , 2 nd	Density-based	Normalized cuts		Spectral Opt.
Two-step Random Walks [82]	2 nd	All	Normalized cuts		Spectral Opt.
Local Partitioning [87]	2 nd	Density-based	Conductance	Local Method	Spectral Opt.
Message Passing [85]	2 nd	All		Affinity propagation	$\mathcal{O}(In C + 2n^2)$
Directed Clique Percolation [88]	2 nd	Density-based	k -clique		$\mathcal{O}(\exp(n))$
Local Density [89, 90]	2 nd	Density-based	Density	Local method	
Infomap [18]	3 rd	Pattern-based	Code length	Compression	Optimization Method
AutoPart [93]	3 rd	Density-based	Code length	Compression	$\mathcal{O}(m C ^2)$
Mixture Models [94, 95, 96]	3 rd	All		Mixture Models, EM	Convergence of EM
Blockmodels [101, 103, 102, 104]	3 rd	Density-based			Parameter Estimation
Co-Blockmodel [105]	3 rd	Density-based		Spectral	Spectral decomposition
Community Kernels [106]	3 rd	All		Kernel's Score	Linear approximation

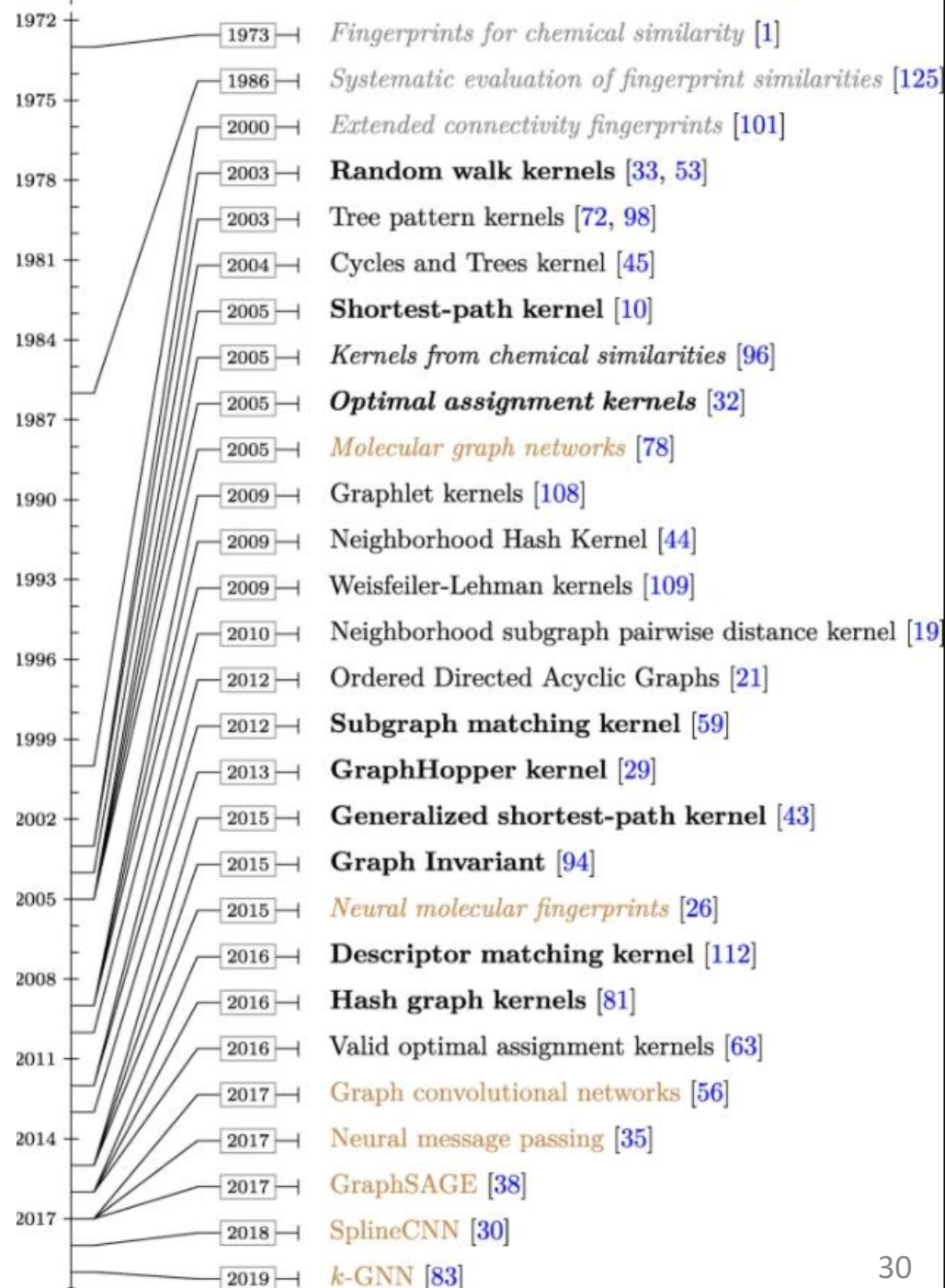
Mining tasks: Graph classification

- Label graphs or elements (i.e., vertices, links, or subgraphs) in a graph.



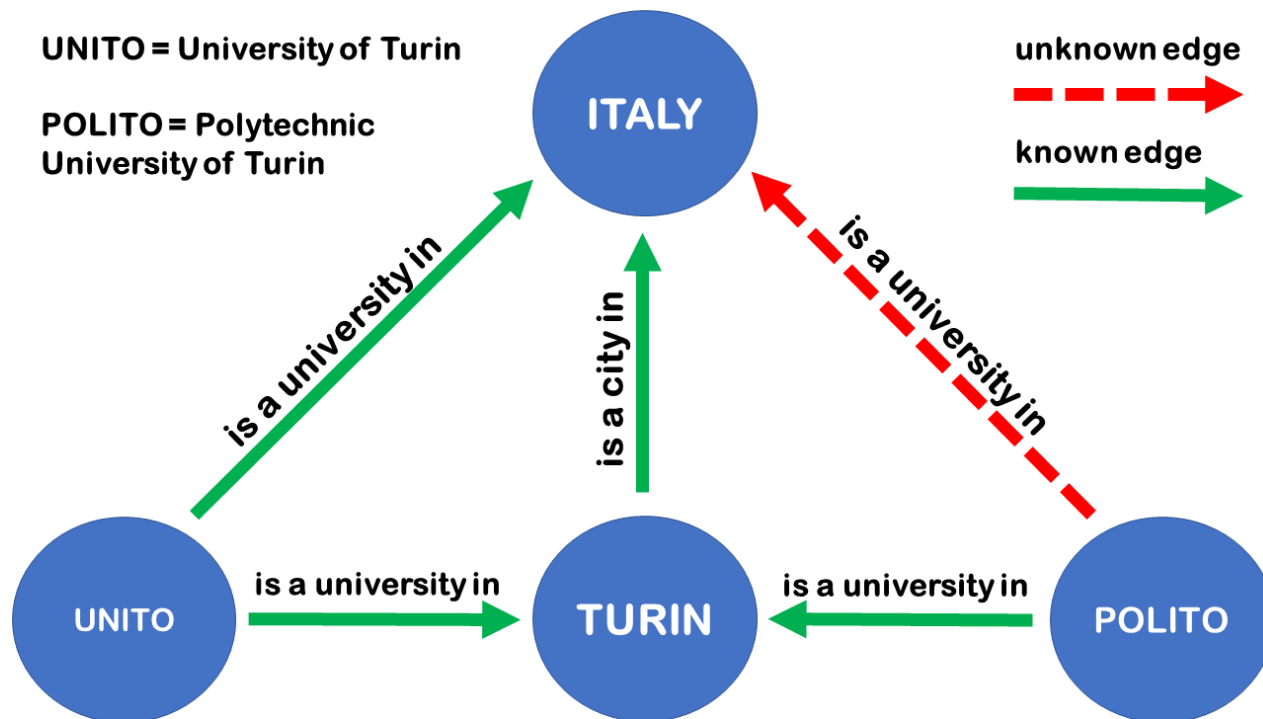
Classify a transaction / user as fraud or non-fraud

Mining tasks: Graph classification



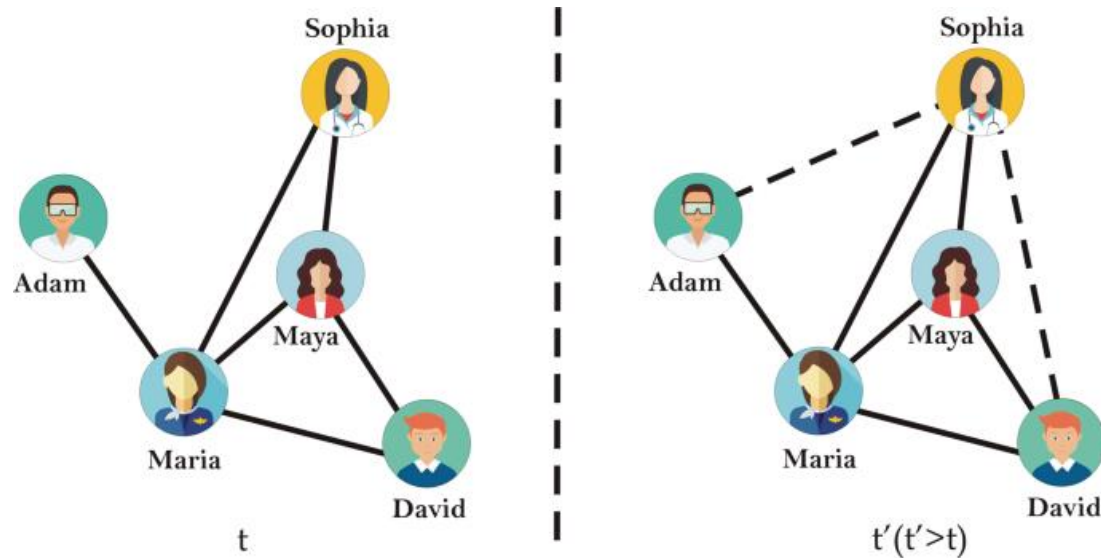
Mining tasks: Link prediction

- Predict the existence of a link between two entities.

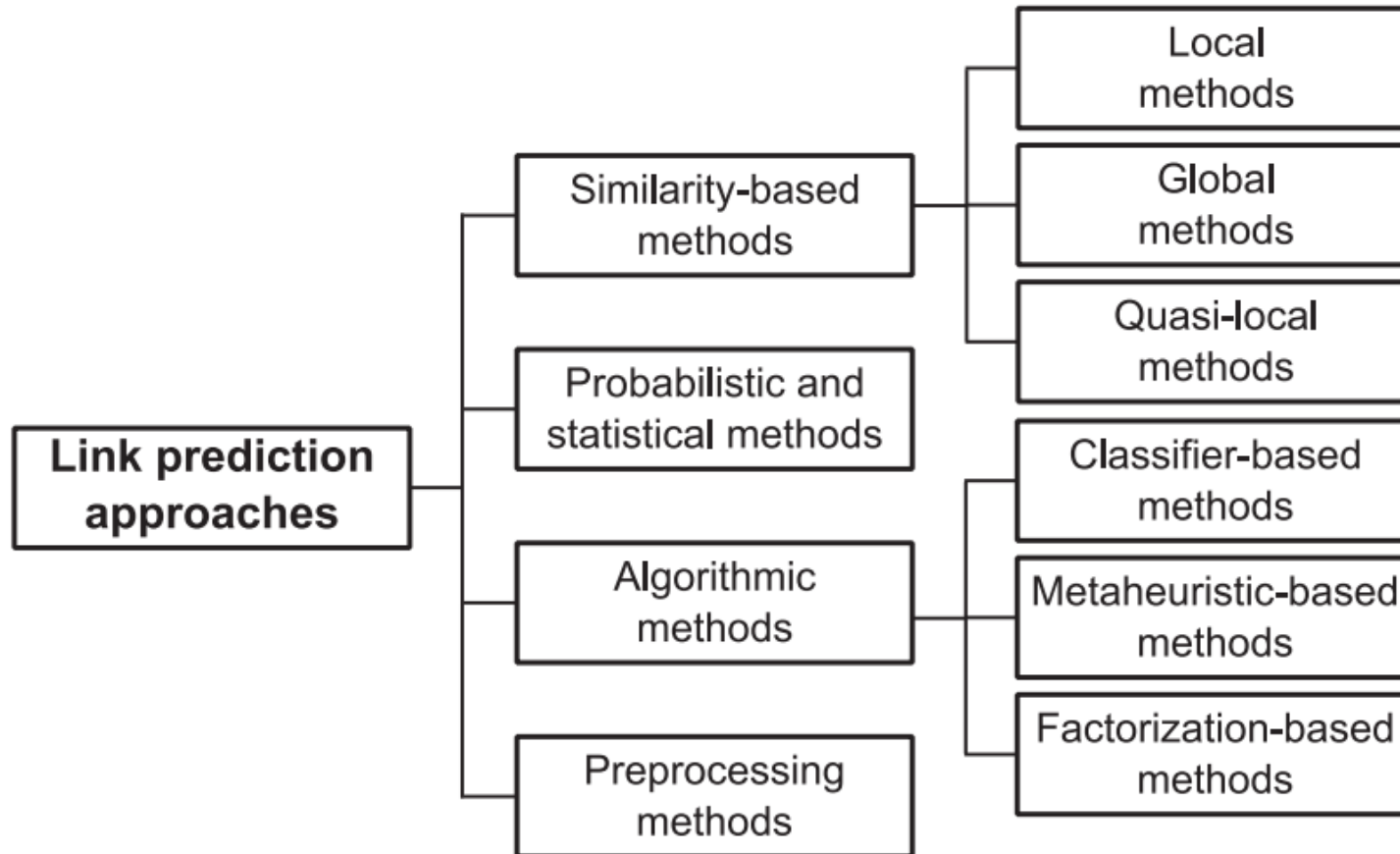


Mining tasks: Link prediction

- Prediction can be performed either **statically** or **over time**.



Mining tasks: Link prediction

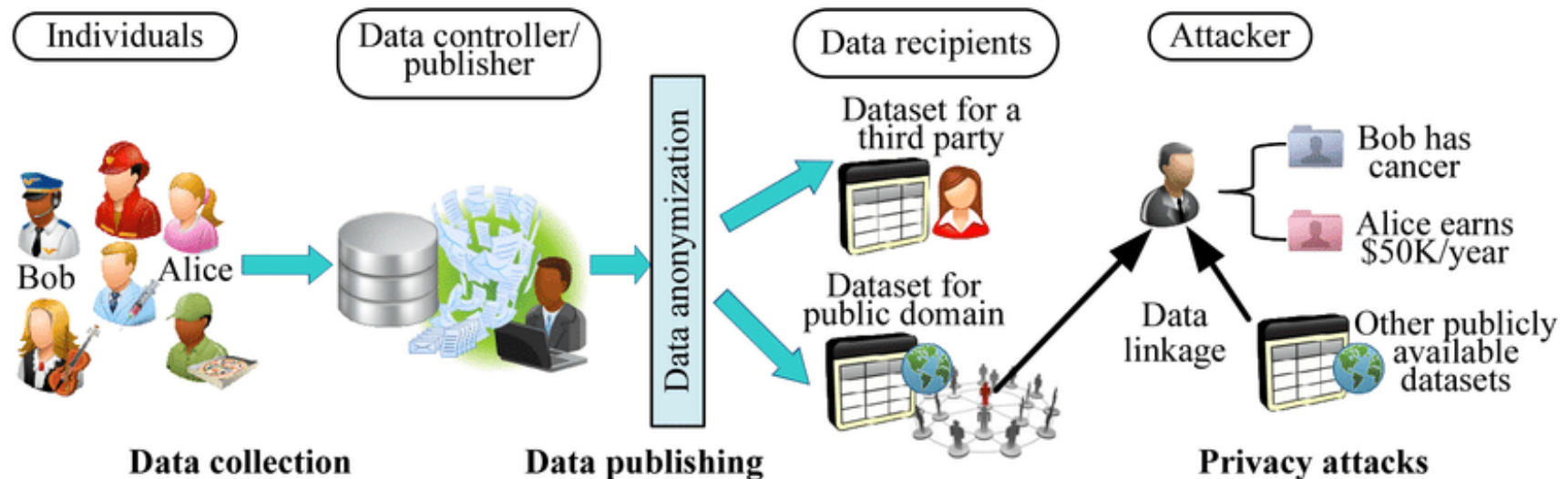


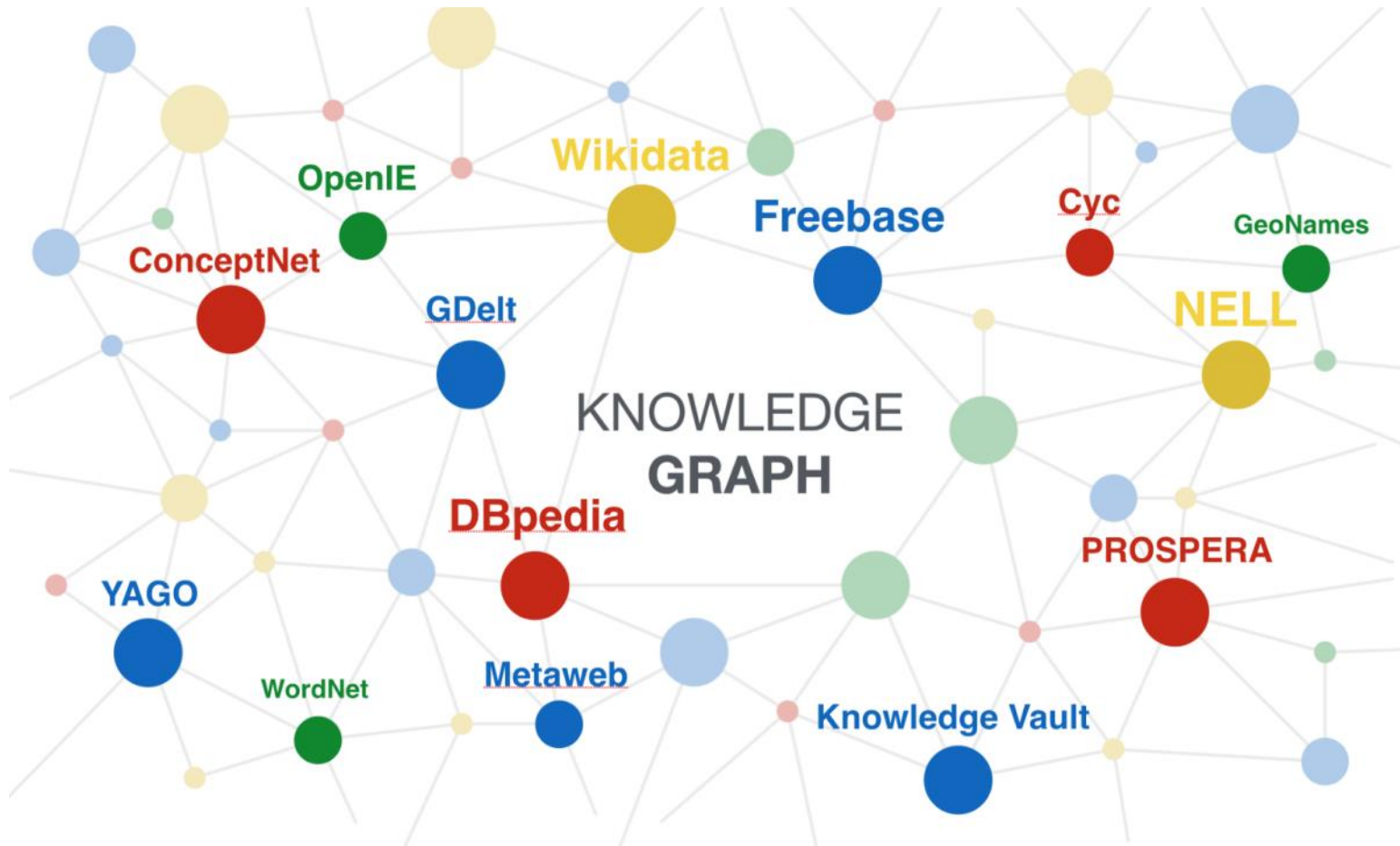
Mining tasks: Link prediction

Type	Name	Time Complexity	Reference
Local	CN	$O(vk^3)$	Liben-Nowell and Kleinberg [2007]
	AA	$O(vk^3)$	Adamic and Adar [2003]
	RA	$O(vk^3)$	Zhou et al. [2009]
	RA-CNI	$O(vk^4)$	Zhang et al. [2014]
	LLHN	$O(vk^3)$	Leicht et al. [2006]
	IA1	$O(vk^4)$	Dong et al. [2011]
	IA2	$O(vk^3)$	Dong et al. [2011]
	MI	$O(nk^6)$	Tan et al. [2014]
	LNB	$O(O(f(z)) + vk^3)$	Liu et al. [2011]
	CAR	$O(vk^4)$	Cannistraci et al. [2013]
Global	NSP	$O(ev \log v)$	Liben-Nowell [2005]
	RWR	$O(cv^2k)$	Tong et al. [2006]
	FP	$O(cv^2k)$	Vanunu and Sharan [2008]
	MERW	$O(cv^2k)$	Li et al. [2011]
	PLM	$O(v^3)$	Fouss et al. [2007]
	ACT	$O(v^3)$	Fouss et al. [2007]
Quasi-local	LPI	$O(lv^2k)$	Lü et al. [2009]
	LRW	$O(lv^2k)$	Liu and Lü [2010]
	SRW	$O(lv^2k)$	Liu and Lü [2010]
	ORA-CNI	$O(vk^6)$	Zhang et al. [2014]
	FL	$O(lv^2k)$	Papadimitriou et al. [2012]
	PFP	$O(vlk^l)$	Lichtenwalter et al. [2010]

Mining tasks: Privacy-preserving

- Removed identity information can still be recovered through linkage.
- How to accomplish privacy-preserving without violating the graph's overall structure?





Data sources and Tools

Graph data sources

[Stanford Network Analysis Platform \(SNAP\)](#)

- A wide variety of graphs representing social networks, citation networks, online communities, and more



[Wikidata](#)

- A collaboratively edited multilingual knowledge graph of 96,886,330 data items

[Yet Another Great Ontology \(YAGO\)](#)

- Augment WordNet with common knowledge facts from Wikipedia
- Originally consisted of more than 1 million entities and 5 million facts



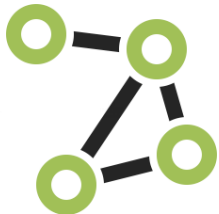
Graph data sources

Open Graph Benchmark

- A collection of realistic, large-scale, and diverse benchmark datasets for machine learning on graphs



TUDatasets



- A collection of benchmark datasets for the evaluation of graph kernels and graph neural networks

Freebase 15K (FB15k)

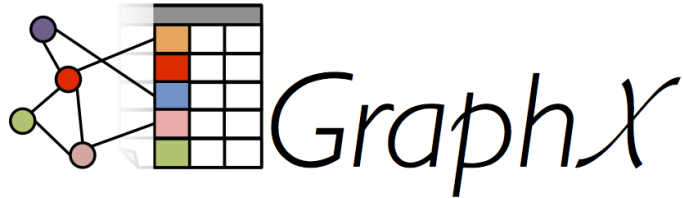
- A collection of 592,213 triplets with 14,951 entities and 1,345 relationships.

Graph databases

- They are NoSQL databases that use graph structures to describe and store data.
- Semantic queries can be done efficiently on nodes, edges, and their properties.



Graph tools



Spark API for parallel computation on graphs

- A collection of algorithms and builders to simplify graph analytic tasks
- Available only in Scala and RDD
- GraphFrames package extends GraphX to Python and DataFrames
- Alternatives: Apache Giraph, Pregel (Google)



plotly

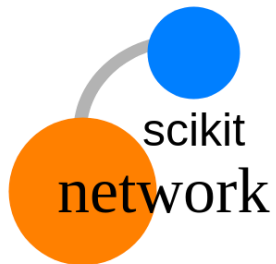
• A tool for graphing, analytics, and statistics

- Available for Python, R, MATLAB, Perl, Julia, Arduino, and REST
- Plotly for Python is free and under the MIT license.

Graph tools

- A free Python library for studying graphs and networks

- Create, manipulate, and study of the structure, dynamics, and functions of complex networks



- A free Python library for of large graphs

- Memory-efficient representation as sparse matrices in the CSR format of scipy
 - Fast algorithms, simple API inspired by scikit-learn

...the end.

